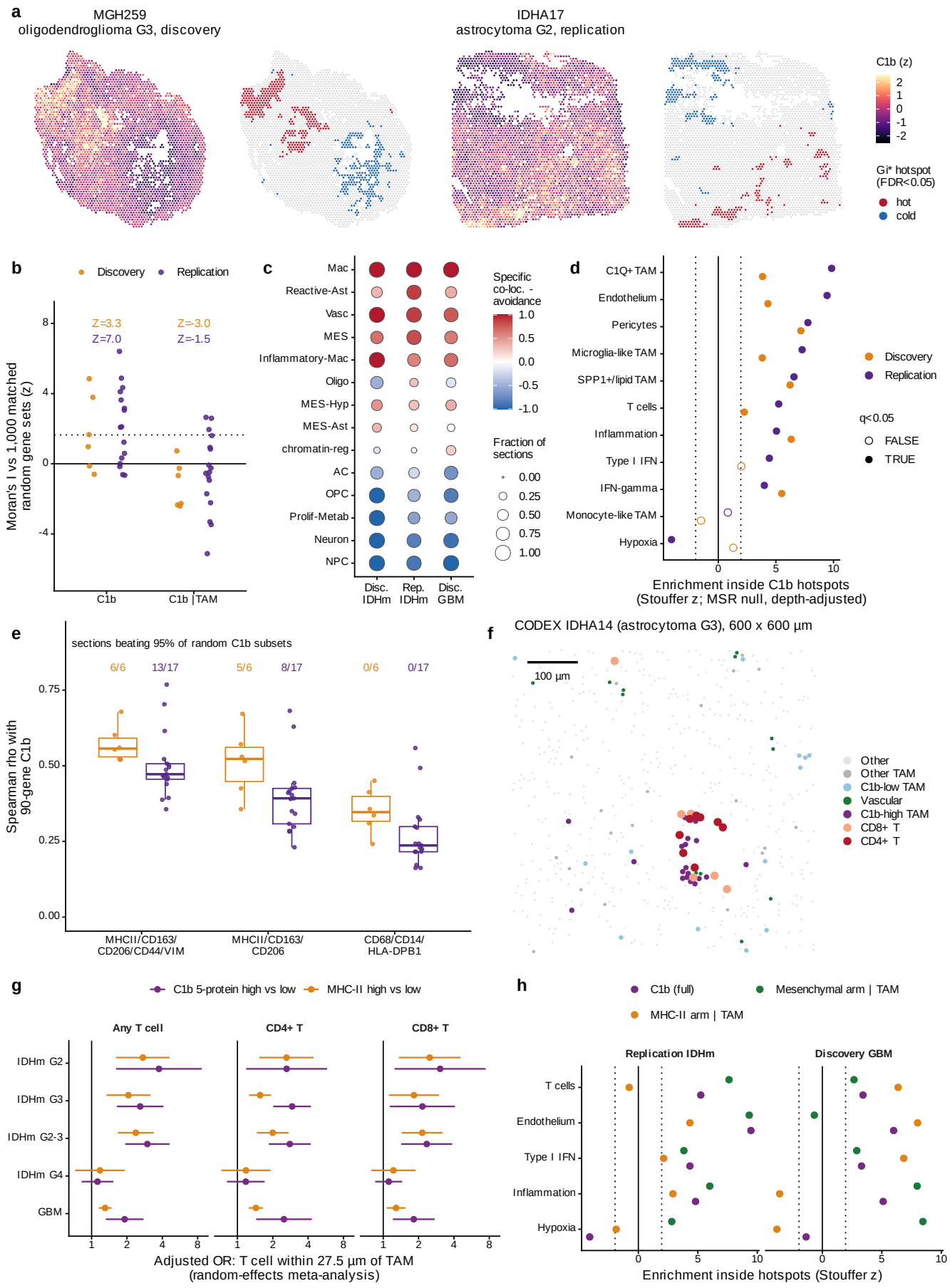

a
MGH259
oligodendroglioma G3, discovery
IDHA17
astrocytoma G2, replication
C1b (z)
2
1
0
-1
-2
Gi* hotspot
(FDR<0.05)
hot
cold
b
Discovery
Replication
Z=3.3
Z=7.0
Z=-3.0
Z=-1.5
Moran's I vs 1,000 matched
random gene sets (z)
C1b
C1b | TAM
c
Mac
Reactive-Ast
Vasc
MES
Inflammatory-Mac
Oligo
MES-Hyp
MES-Ast
chromatin-reg
AC
OPC
Prolif-Metab
Neuron
NPC
Disc.
IDHm
Rep.
IDHm
Disc.
GBM
Specific
co-loc. -
avoidance
1.0
0.5
0.0
-0.5
-1.0
Fraction of
sections
0.00
0.25
0.50
0.75
1.00
d
C1Q+ TAM
Endothelium
Pericytes
Microglia-like TAM
SPP1+/lipid TAM
T cells
Inflammation
Type I IFN
IFN-gamma
Monocyte-like TAM
Hypoxia
Enrichment inside C1b hotspots
(Stouffer z; MSR null, depth-adjusted)
Discovery
Replication
q<0.05
FALSE
TRUE
e
sections beating 95% of random C1b subsets
6/6
13/17
5/6
8/17
0/6
0/17
Spearman rho with
90-gene C1b
MHCII/CD163/
CD206/CD44/VIM
MHCII/CD163/
CD206
CD68/CD14/
HLA-DPB1
f
CODEX IDHA14 (astrocytoma G3), 600 x 600 µm
100 µm
Other
Other TAM
C1b-low TAM
Vascular
C1b-high TAM
CD8+ T
CD4+ T
g
C1b 5-protein high vs low
MHC-II high vs low
Any T cell
CD4+ T
CD8+ T
IDHm G2
IDHm G3
IDHm G2-3
IDHm G4
GBM
Adjusted OR: T cell within 27.5 µm of TAM
(random-effects meta-analysis)
h
C1b (full)
Mesenchymal arm | TAM
MHC-II arm | TAM
Replication IDHm
Discovery GBM
T cells
Endothelium
Type I IFN
Inflammation
Hypoxia
Enrichment inside hotspots (Stouffer z)